German University in Cairo GUC
Faculty of Media Engineering and Technology
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Eng. Dinah Waref

DMET1067—Deep Learning for Computer Vision
Spring Semester 2026

DLCV Assignment

Due: Friday, 17.04.2026 at 11:59 pm.

This project aims to build an emotion classification system from scratch. The core task is to classify the emotions within each frame based on the facial expression. You will begin by designing and constructing a deep learning model from scratch for emotion classification. This involves defining the network architecture and implementing the training process. Finally, you will test the model's performance on unseen data to evaluate its accuracy in classifying emotions.

Teams: You will be working in teams of five to **six**. One student per team (Spokesperson) should submit your team details using the following form: https://forms.gle/iShPXqsnprzAr5WNA. The deadline for team submission is ***Friday, 20 March 2026.***

Part 1: (70%)

In this assignment, you are tasked to build models to classify the images into one of the defined classes. You will learn to create a Convolutional Network from scratch *(without the use of any external libraries)* in the first model. In the 2nd model, you can use ML libraries to create and train CNN for image classification.

The model will be trained on four classes of your choice from the shared dataset you collectively created.

Data Preparation: (5%)

1- Use one of the publicly available benchmark datasets to classify emotions.

2- Download the images and store the images of each class in separate folders.

3- Split the data for the selected classes into training/validation and testing with ratios of 70/20/10. (2%)

4- Remove any redundant images. (1%)

5- Resize the images to be 512x512x3. (2%)

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First Model: (35%)

Create a convolution network from scratch for feature extraction. Then use the extracted feature vector with a simple unsupervised algorithm for data clustering. To avoid the need for the backpropagation algorithm, we will use predefined filters for the convolution layers.

1- Create a single convolution layer using 5 predefined 3x3x3 filters. Input an image of 512x512x3. (10%)

a. Create a class ConvLayer which has at least two initialization methods

i. Given the number and the size of the filters. This method should create random filter weights.

ii. Given the number and the size of filters and a matrix in 3D that has the filter weights.

b. Use the following filters to initialize an object of the class ConvLayer:

1	1	1
1	1	1
1	1	1

(a)

0	0	0
0	1	0
0	0	0

(b)

-1	0	1
-2	0	2
-1	0	1

(c)

-1	-2	-1
0	0	0
1	2	1

(d)

0	-1	0
-1	5	-1
0	-1	0

(e)

2- Create a class PoolingLayer which is initialized by the size of the pooling filter (default 2x2) and the pooling type is either (default) MAX or AVERAGE. (5%)

3- For both the ConvLayer and PoolingLayer create methods to (8%)

i. Iterate the filters on the entire input image/ matrix.

ii. Run a forward pass methods

4- Pass the output of the pooling layer to a simple activation function defined as (2%)

$$\sigma(z) = \begin{cases} z & z \geq 0 \\ 0 & otherwise \end{cases}$$

5- Use the above-created Convolution block to create a network as: (8%)

i. 3 Convolution blocks

ii. Flattening layer

iii. Downsample the output of the flattening layer to the size 1x128.

6- Use the k-means algorithm to classify the images into different classes. (2%)

Second Model: (30%)

Use your favorite DL libraries to create a classification CNN with the following architecture conditions:

1- Build a CNN with 3 convolutional layers, each of which uses ReLU as an activation function, followed by a 2x2 max-pooling filter. Use the default stride for the max-pooling filters. (suggested) (8%)

2- The filter sizes are 3x3, 3x3, 3x3, 5x5, and 7x7, and use 32, 64, 64, 32, and 16 filters for the successive convolution layers, respectively. Use valid spatial convolution only. (suggested) (6%)

3- The convolution blocks are followed by a flattening operation. (3%)

4- Use the flat vector as an input to fully connected layers with only one hidden layer. Use the Sigmoid as the activation function for this layer. (6%)

5- Use the Softmax function for the output layer. (4%)

6- All other hyper-parameters than what has been mentioned here are left to be determined by the developing team. (3%)

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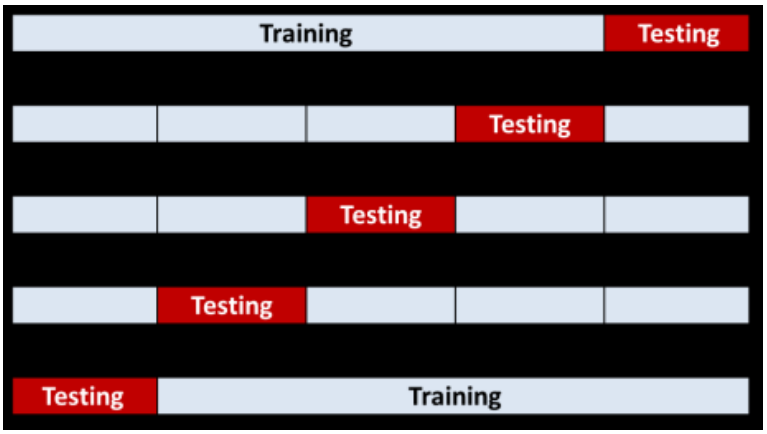
Part 2: (30%)

This last part of the project aims to finalize and report the project outcomes.

Therefore you are requested to complete the following tasks:

1- Build a curve for the accuracy vs the number of iterations. (5%)

2- Perform K-fold cross-validation and report the resulted accuracy and the average of the recorded K-accuracies. The minimum chosen value for K is 4. (5%)



3- Compute and report the confusion matrix. (5%)

Multi Class Confusion Matrix		Predicted				Accuracy
		Class 1	Class 2	...	Class N	
Actual	Class 1	M_{11}	M_{12}	...	M_{1N}	
	Class 2	M_{21}	M_{22}	...	M_{22}	
	...	...	...	...	...	
	Class N	M_{N1}	M_{N2}	...	M_{NN}	
Reliability						*Overall Accuracy*

- Several important information can be observed from the confusion matrix such as:

1- Accuracy of a class: $ACx = (x,x)/ \sum_y (x,y)$

2- Overall Accuracy: $AC = \sum_x (x,x)/ \sum_x \sum_y (x,y)$

3- Overall Error rate: $ER = \sum_x \sum_{y \neq x} (x,y)/ \sum_x \sum_y (x,y)$

1. The entries in the main diagonal represent the number of correct classified pixels, for example (1,1) represent the number of pixels classified as being belong to class 1 while they were actually belonging to class 1.
2. The entries of the form (x,y) such that $x \neq y$, represent the number of pixels which were wrongly classified to belong to class y, while they are actually belong to class x.

4- Build a block diagram for your network architecture. Add in bullet points all chosen values for the hyperparameters. (5%)

5- Report any steps of preprocessing or post-processing of the data. (5%)

6- Compile your work report, results, analysis, and discussion in one Word document or PowerPoint presentation. (5%)

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Assignment Delivery:

All deliverables must be compressed together in one (*.zip) file. Name the file as T_DLCV_Assign.zip where T is the team name. Upload the zip file to Google Drive and submit the *shareable* link to the following form by ***Friday, 17.04.2026 at 11:59 PM***: https://forms.gle/CHfkHnTiEt77NB767

1- Submit a Jupiter notebook file or .py for your well-commented code. Specify in clear text the selected hyper-parameters and any used libraries.

2- A PowerPoint slide or report for the visualization of the created model architecture, annotated with the dimensions of the layers and in/out data. Present the selected classes with some samples of the inputs and the obtained results in the same file. Showing all the visualizations and graphs you created, as well as an explanation of them. Add one slide for the model hyperparameters, the number of training iterations, and any additional notes.

3- A textfile (*.txt) with the team information, including the names and contacts of the team members.

4- Please note that plagiarism or AI-generated codes will result in a zero grade

5- Evaluations will be held, and part of your grade depends on your performance in the set evaluations.

Bonus: (3%)

We're giving a 1.5% bonus for each of the extra tasks below completed, but there's a limit of 3% in total, meaning if you do one bonus task you get 1.5% and if you do two you get 3% which is the maximum.

1. Data Augmentation: Implement data augmentation techniques such as rotation, flipping, and cropping to artificially increase the size of your dataset and improve the generalization of your models.
2. Model Regularization: Experiment with different regularization techniques such as L1/L2 regularization, dropout, and batch normalization to prevent overfitting and improve the generalization of your models.
3. Hyperparameter Tuning: Conduct a hyperparameter tuning process using techniques such as grid search or random search to find the optimal values for parameters such as learning rate, batch size, and number of filters in the convolutional layers.
4. Transfer Learning: Explore the use of pre-trained convolutional neural network models (e.g., VGG, ResNet, or Inception) as feature extractors and fine-tune them on your dataset to potentially improve classification performance.
5. Advanced Evaluation Metrics: In addition to accuracy, compute other evaluation metrics such as precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC-ROC) to gain a more comprehensive understanding of your model's performance.
6. Model Deployment: Investigate methods for deploying your trained models into production environments, such as converting them to optimized formats (e.g., TensorFlow Lite or ONNX) for inference on mobile or edge devices.
7. Error Analysis: Perform an in-depth analysis of the misclassified examples to identify common patterns or challenges that your models struggle with, which could provide insights for future improvements.